

Clinical-Audit-Automation

Python-based tool for Medicare billing compliance and clinical data integrity.

🏠 Project Overview

This project was inspired by my four-year tenure as a Clinical Coordinator and Supervisor at ****Specsavers****. During my time there, I managed complex Medicare billing cycles and health fund audits. I developed this script to demonstrate how automation can replace manual record-checking to ensure 100% data accuracy in a clinical setting.

🌐 Why This Matters for Biomedical Informatics

Informatics is the bridge between clinical domain expertise and computational efficiency. This tool reflects that bridge by:

- ****Ensuring Data Integrity:**** Identifying missing or incorrect billing codes.
- ****Scaling Clinical Oversight:**** Moving from manual "spot-checks" to automated full-dataset validation.
- ****Operational Optimization:**** Reducing human error in high-volume healthcare environments.

🚀 How It Works

The script uses the `pandas` library to process a mock clinical dataset. It executes a validation engine that:

1. Cross-references patient eligibility status against submitted billing codes.
2. Flags discrepancies where required information is missing.
3. Outputs a clean "Audit Report" for administrative review.

🛠️ Technical Stack

- ****Language:**** Python 3.x
- ****Libraries:**** Pandas (Data manipulation)

📁 Repository Structure

- `audit\_tool.py`: The core Python script containing the audit logic.
- `mock\_data.csv`: Sample data representing patient appointments and billing statuses.
- `About\_The\_Code.pdf`: A detailed explanation of the project's logic and clinical context.

📧 Contact

Hamna